		$(\frac{T_{new}}{T})^2 = (\frac{R/30}{R})^3 \Rightarrow R = 0.166 \text{ day} \approx 4.0 \text{ hour}$
14	**A**	Conservation of energy, $E_{Final} = E_{Initial}$ $\frac{1}{2}mv^2 - \frac{GMm}{R} = 0 - \frac{GMm}{3R}$ $v = \sqrt{\frac{4GM}{3R}}$